



CSRS
Centre Suisse de Recherches
Scientifiques en Côte d'Ivoire



Sequencing Data Quality Control

From raw reads to assembly quality assessment

Molecular and Genomic Approaches: Bacterial Meningitis
Diagnosis and Surveillance in Africa

28 September - 2 October 2026 | CSRS, Côte d'Ivoire

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Can you see the details?



Low quality: what breed? Eye colour? Collar?

Optimise the quality and the duration of the analyses

Can you see the details?

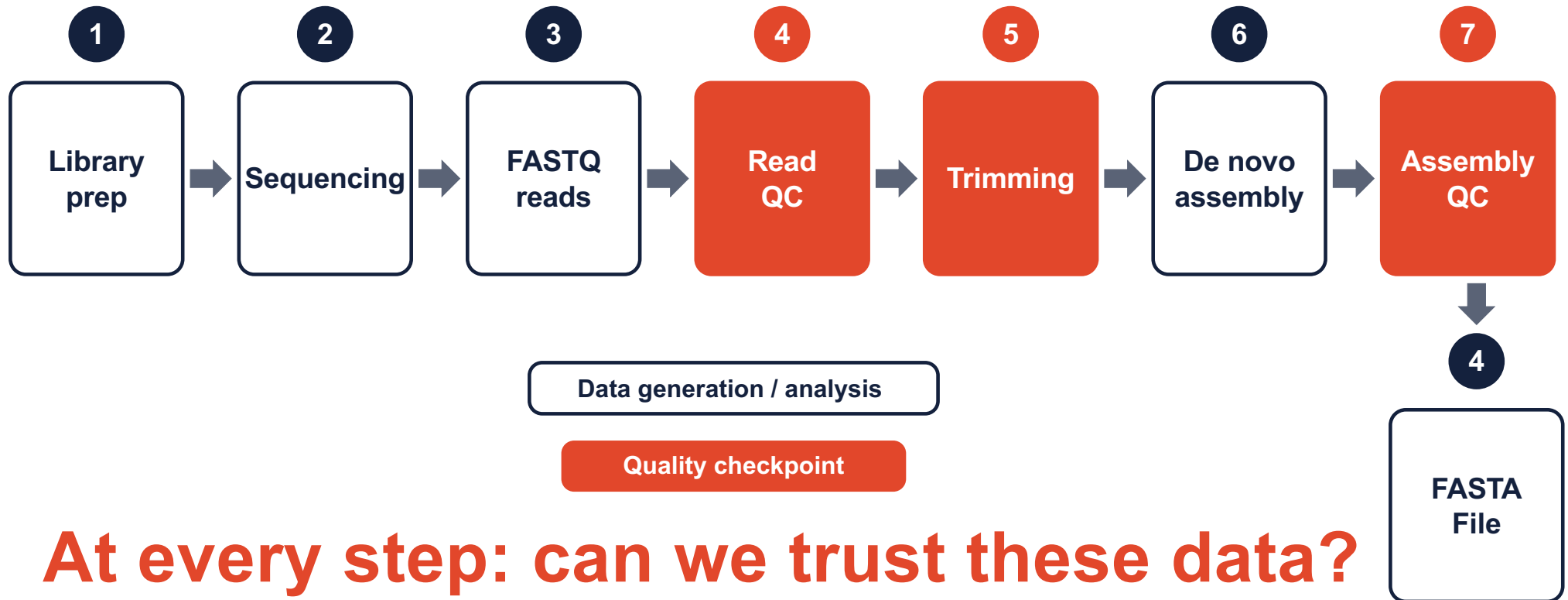


High quality: did you spot everything?

Avoid wrong biological interpretations

WELCOME

Our roadmap today



At every step: can we trust these data?

WARM-UP

Warm-up: quick show of hands

After a sequencing run, what do you receive from the sequencer?

A

A FASTA file of the genome

B

FASTQ files of reads

C

A phylogenetic tree

D

A table of resistance genes

Output fastq file organization

- Reads are automatically demultiplexed during basecalling
- Fastq files are generated in real-time (every 10 minutes in this example), so multiple fastq files are generated for each barcode (sample)

 barcode_al...379472e5.tsv 	 barcode01  >	 FAY32108_p...5_0.fastq.gz  Aug 13, 2024 at 1:06 PM
 fastq_fail  >	 barcode02  >	 FAY32108_p...5_1.fastq.gz  Aug 13, 2024 at 1:16 PM
 fastq_pass  >	 barcode03  >	 FAY32108_p...5_2.fastq.gz  Aug 13, 2024 at 1:26 PM
 final_summa...79472e5.txt 	 barcode04  >	 FAY32108_p...5_3.fastq.gz  Aug 13, 2024 at 1:36 PM
 other_reports  >	 barcode05  >	 FAY32108_p...5_4.fastq.gz  Aug 13, 2024 at 1:46 PM
 pod5  >	 barcode06  >	 FAY32108_p...5_5.fastq.gz  Aug 13, 2024 at 1:56 PM
 pore_activity...9472e5.csv 	 barcode07  >	 FAY32108_p...5_6.fastq.gz  Aug 13, 2024 at 2:06 PM
 report_FAY3...4b3362.html 	 barcode08  >	 FAY32108_p...5_7.fastq.gz  Aug 13, 2024 at 2:16 PM
 report_FAY3...4b3362.json 	 barcode09  >	 FAY32108_p...5_8.fastq.gz  Aug 13, 2024 at 2:26 PM
 report_FAY3...a4b3362.md 	 barcode10  >	 FAY32108_p...5_9.fastq.gz  Aug 13, 2024 at 2:36 PM
 sample_she...a4b3362.csv 	 barcode11  >	 FAY32108_p...10.fastq.gz  Aug 13, 2024 at 2:46 PM
 sequencing...379472e5.txt 	 barcode12  >	 FAY32108_p...5_11.fastq.gz  Aug 13, 2024 at 2:56 PM
 throughput_...79472e5.csv 	 barcode13  >	 FAY32108_p...12.fastq.gz  Aug 13, 2024 at 3:06 PM
	 barcode14  >	 FAY32108_p...13.fastq.gz 
	 barcode15  >	 FAY32108_p...14.fastq.gz 

Sequencing run QC using the “report_***.html” file

^ Run summary

DATA OUTPUT

Estimated bases

22.27 Gb

Reads generated

7.02 M

Estimated N50

8.71 kb

BASECALLING

Reads called

100%

Reads called (min Q score: 9)

6.56 M

459.67 k

Pass

Fail

Bases called (min Q score: 9)

21.22 Gb

920.17 Mb

Pass

Fail

RUN DURATION

Run time

72 hrs 0 mins / 72 hrs 0 mins

Elapsed time



Run limit



Run status

FINISHED · Target runtime has been reached

Cumulative output as a function of time

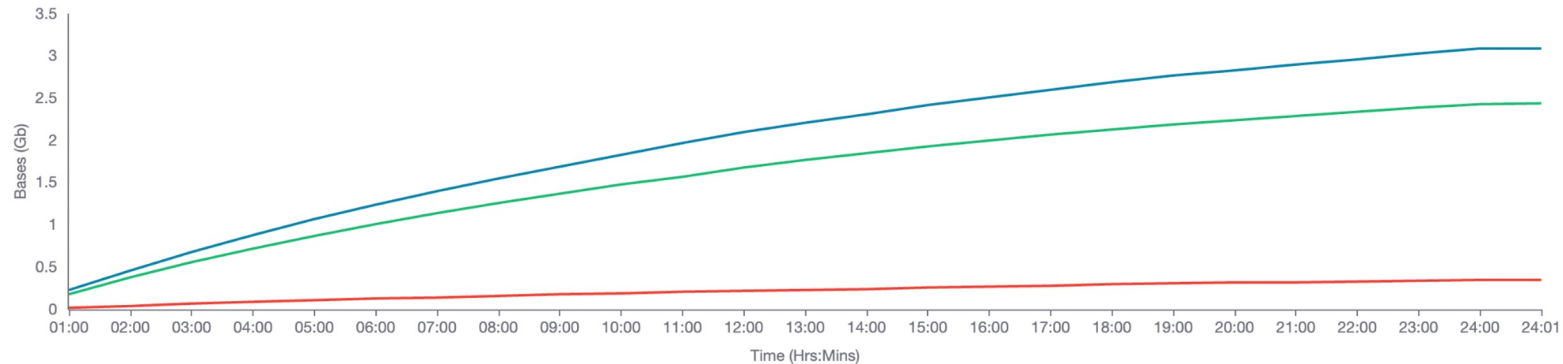
^ CUMULATIVE OUTPUT

The cumulative output shows the total amount of bases or reads sequenced over time by your device.

Bases

Legend

Estimated	Passed	Failed
Predicted total number of bases, prior to basecalling	Bases equal to or above the quality score threshold.	Bases below the quality score threshold.



Reads

Legend

Total	Passed	Failed	Skipped
Total number of reads, including passed, failed and skipped.	Reads equal to or above the quality score threshold.	Reads below the quality score threshold.	Reads that will not be basecalled. Post run basecalling is possible.

Cumulative output as a function of time

^ CUMULATIVE OUTPUT

The cumulative output shows the total amount of bases or reads sequenced over time by your device.

Bases

Legend

— Estimated

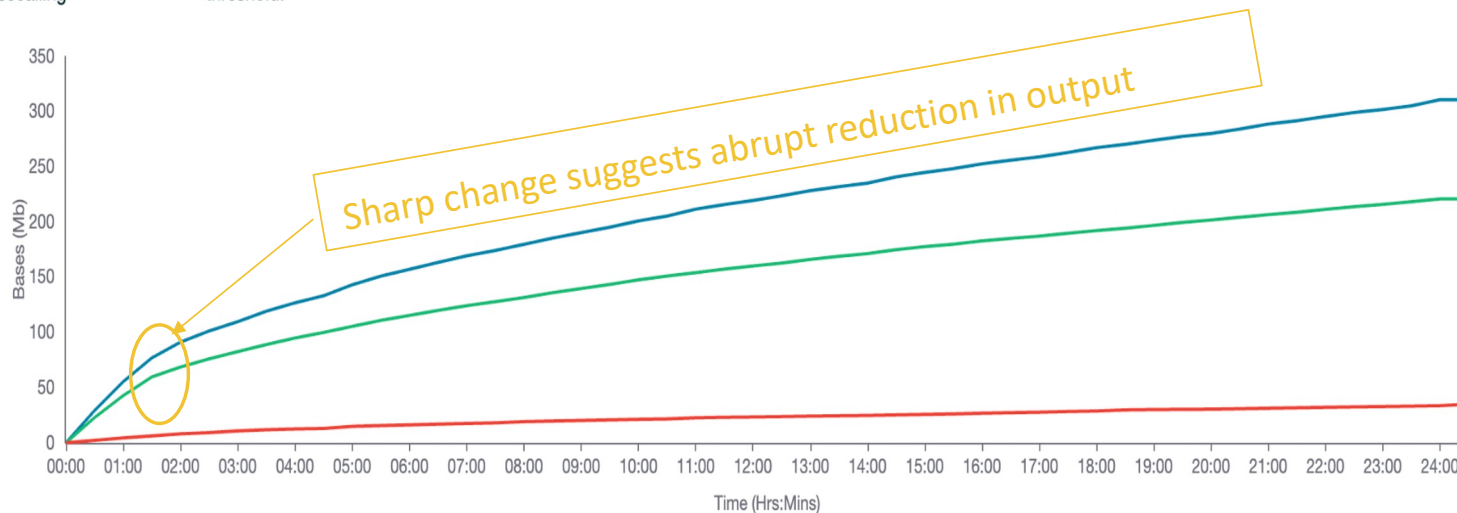
Predicted total number of bases, prior to basecalling

— Passed

Bases equal to or above the quality score threshold.

— Failed

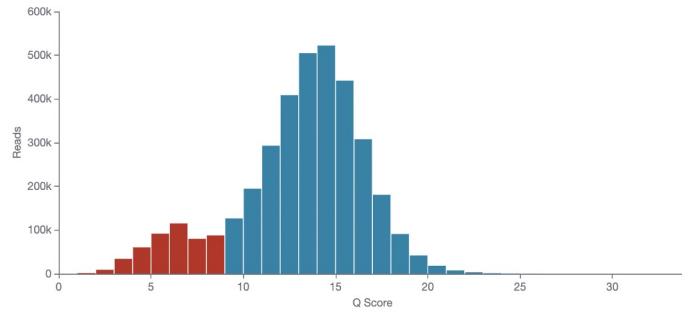
Bases below the quality score threshold.



Q-score

Q Score histogram

Passed simplex reads
Failed simplex reads



Q Score over time

Legend

— Mode

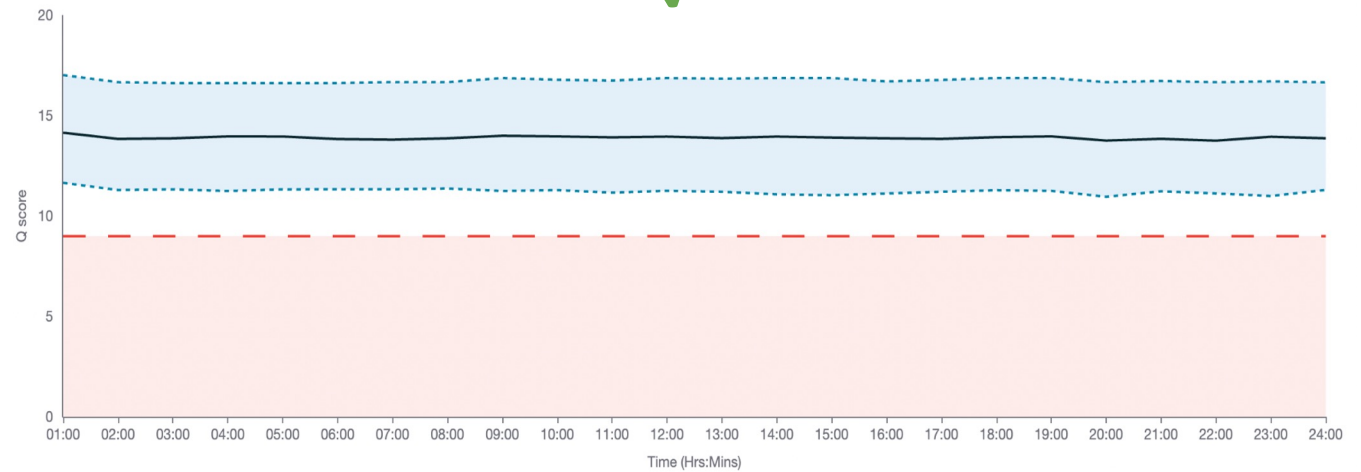
The most frequent quality score of reads in the run.

□ Spread

The spread of quality scores, found by calculating full width half maximum.

--- Min. quality score

Minimum quality score to be accepted as a passed read.



Q-score

^ QUALITY SCORE

The quality score is calculated as basecalling is performed on your device. Reads that fall below the minimum value of 9 will be classified as failed reads. You can alter the accepted minimum quality score in MinKNOW.

Legend

— Mode

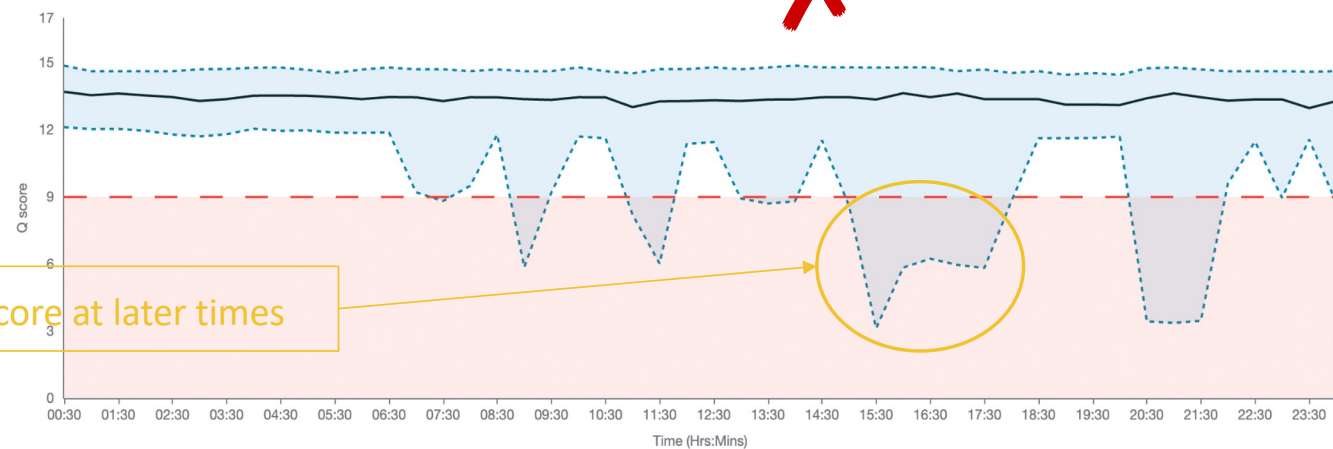
The most frequent quality score of reads in the run.

--- Spread

The spread of quality scores, found by calculating full width half maximum.

--- Min. quality score

Minimum quality score to be accepted as a passed read.



Pore activity over time

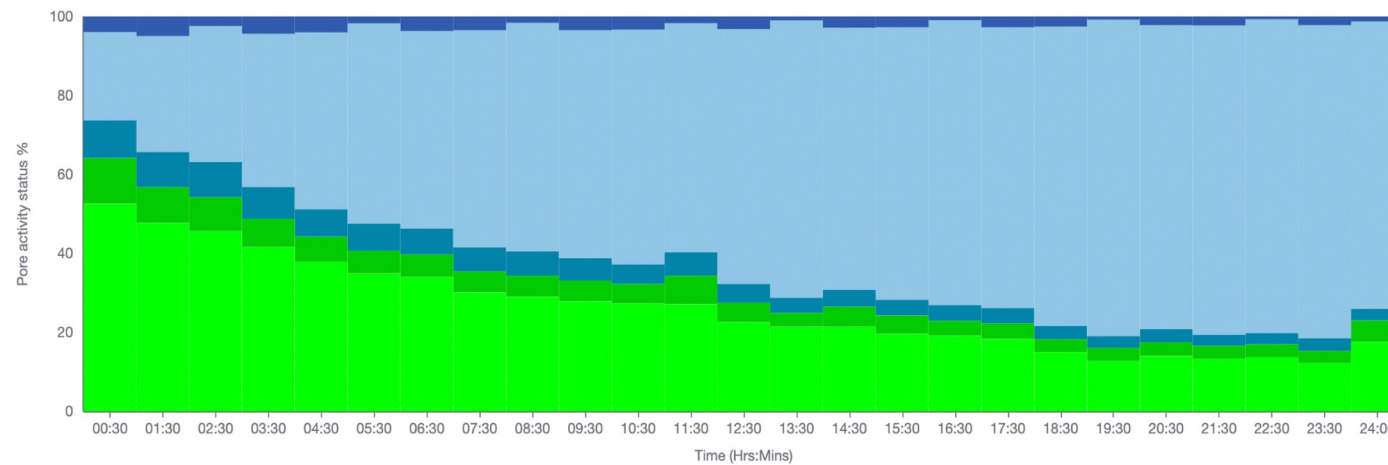
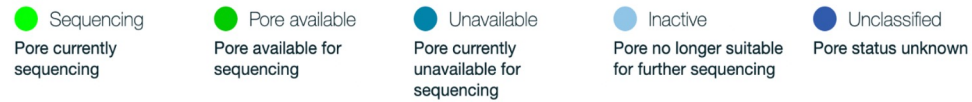
Run health

PORE ACTIVITY

The Pore activity graph shows the performance of your sample as it is being sequenced during a run.

☒ Show grouped

Legend



Pore activity over time

PORE ACTIVITY

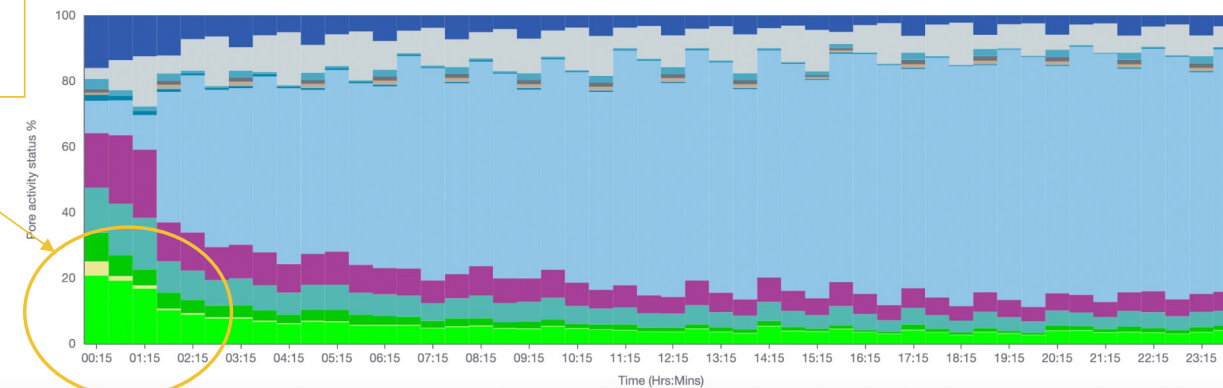
The Pore activity graph shows the performance of your sample as it is being sequenced during a run.

Legend

- Sequencing
Pore currently sequencing
- Adapter
Pore currently sequencing adapter
- Pore available
Pore available for sequencing
- Unavailable
Pore unavailable for sequencing
- Active feedback
Channel ejecting analyte
- No pore
No pore detected in channel
- Out of range-high
Current is positive but unavailable for sequencing
- Out of range-low
Current is negative but unavailable for sequencing
- Multiple
Multiple pores detected. Unavailable for sequencing.
- Saturated
The channel has switched off as current levels exceed hardware limitations
- Zero
Pore currently unavailable for sequencing
- Channel disabled
Channel is disabled and awaiting another pore scan
- Unclassified
Pore status unknown

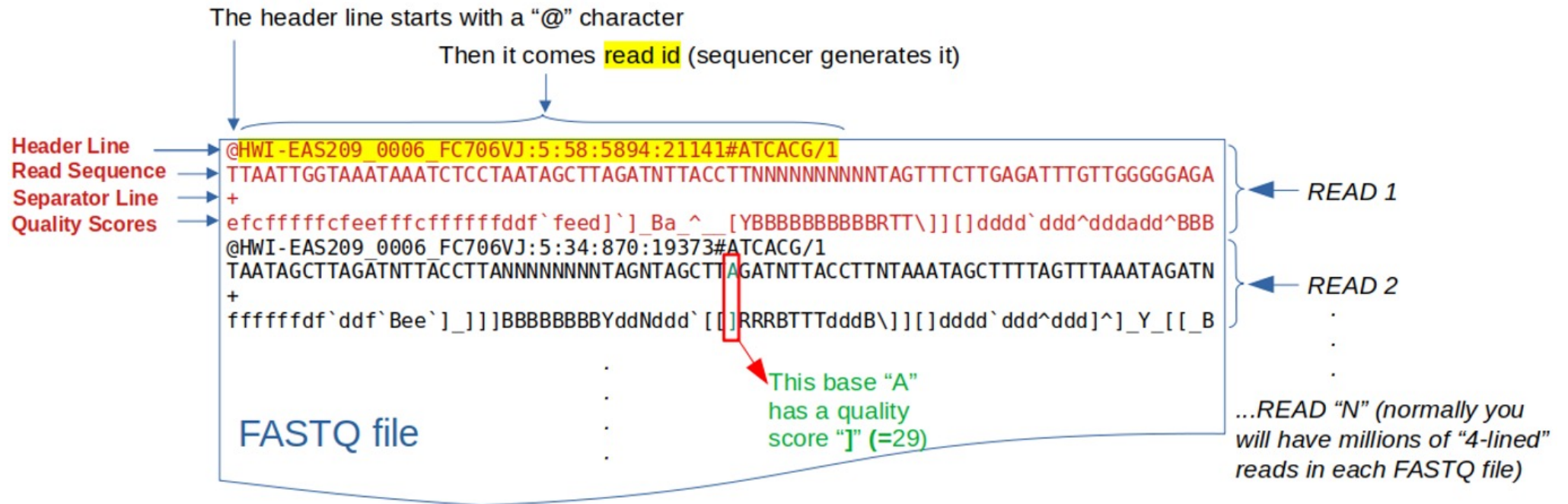
Lots of unusual pore statuses

Low starting % of active pores



The FASTQ file: 4 lines per read

fastq file



Reading the quality string

FASTQ record: 4 lines per read

IDENTIFIER:	@SIM:1:FCX:4:15:6329:1045 1:Y:0:2
SEQUENCE:	TCGNNCTCAACGCCCTGCATANGANAAGACAGAANC
+	+
QUALITY SCORES:	<>;##=><9=AAAAAAAAAA9#:<#<;<<????#<=

Line 4 = quality scores.
One character per base

ASCII code of each quality character

	33	59	64	73		104	126
ASCII code	!"#\$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{ }~						
Sanger	0	26	31	40			
Illumina / Nanopore	0	26	31	41			

Sanger: Q0 to Q40

Illumina / Nanopore: Q0 to Q41

Symbols on the left = low quality. Letters = good quality!

Phred scores: quality as probability

$$Q = -10 \times \log_{10}(P)$$

$$P = 10^{-Q/10}$$

P = probability that the base is wrong

Q30 = 1 error per 1,000 bases

Nanopore scores typically reach up to about Q41

Phred (Q)	Error probability	Base accuracy
Q10	1 in 10	90%
Q20	1 in 100	99%
Q30	1 in 1,000	99.9%
Q40	1 in 10,000	99.99%

FastQC: your first look at the data

- Works directly on FASTQ files
- One report per file: HTML or interactive window
- Each module gets a traffic light
- Flags are guides, not verdicts



Normal



Slightly unusual

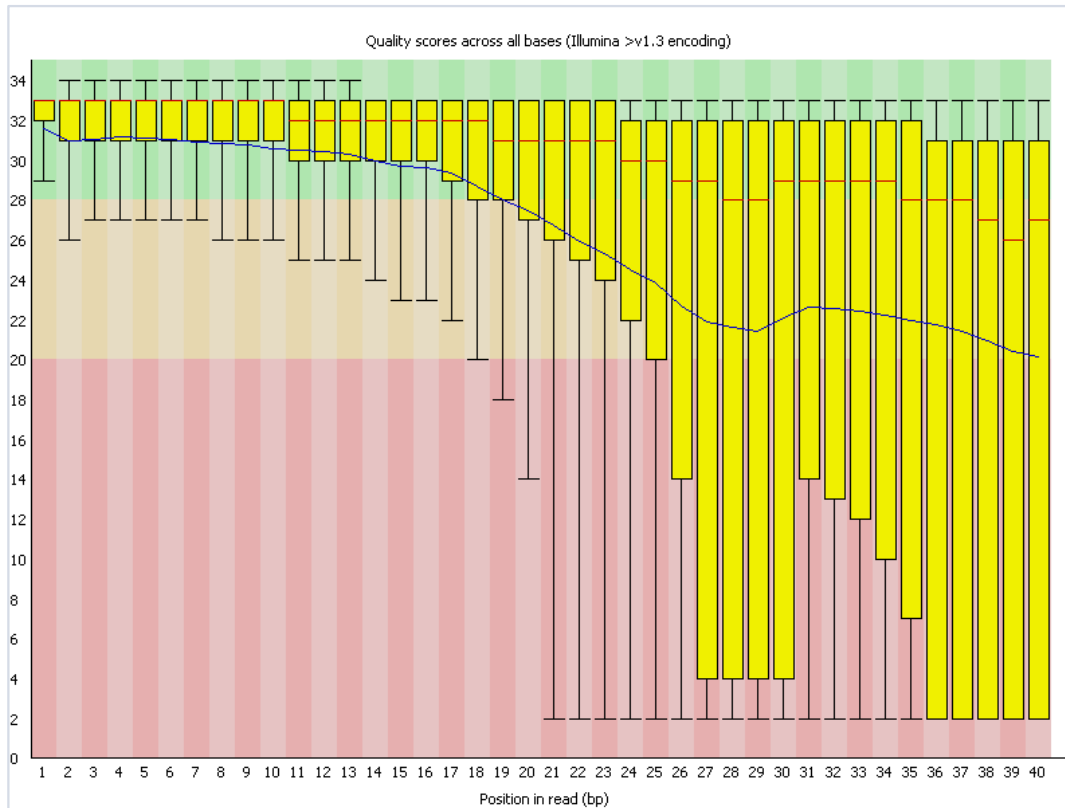


Very unusual

Summary

- ✓ [Basic Statistics](#)
- ! [Per base sequence quality](#)
- ✓ [Per tile sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ✗ [Per base sequence content](#)
- ✗ [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ✓ [Sequence Length Distribution](#)
- ✗ [Sequence Duplication Levels](#)
- ✗ [Overrepresented sequences](#)
- ✗ [Adapter Content](#)

Per base sequence quality



This plot tells you whether to trim.

Per-base quality: warning or fail?



WARNING

Lower quartile < 1 , or median < 9



FAIL

Lower quartile < 5 , or median < 9



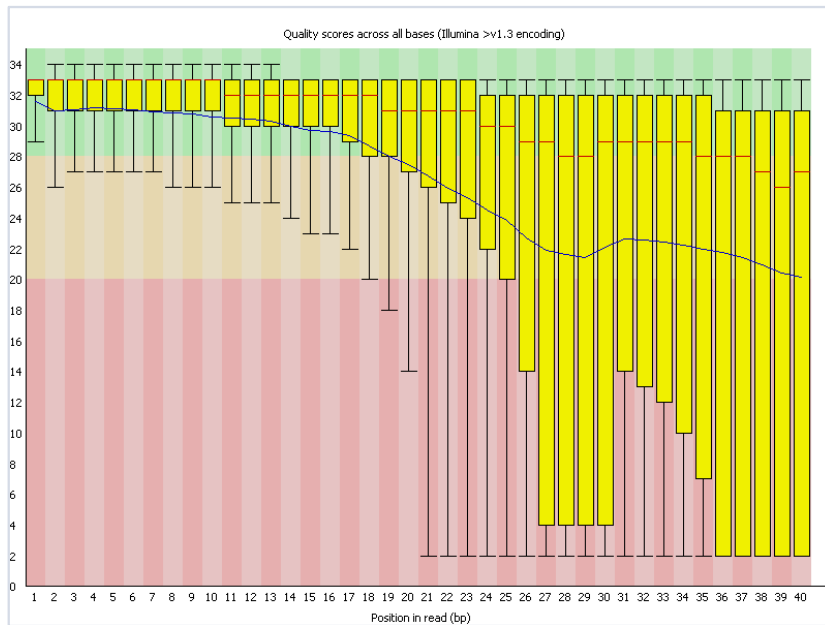
RULE OF THUMB

Median below ~Q20: consider trimming

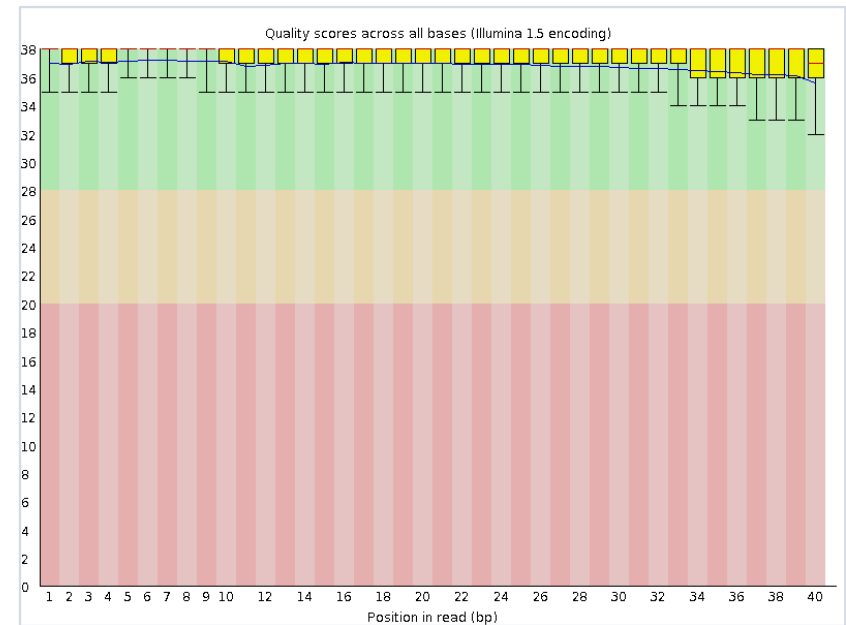
A short early dip that recovers can be a transient run problem (bubbles)

Variable-length reads: low coverage at the far end can trigger flags. Check the length distribution

Good vs bad: per base quality



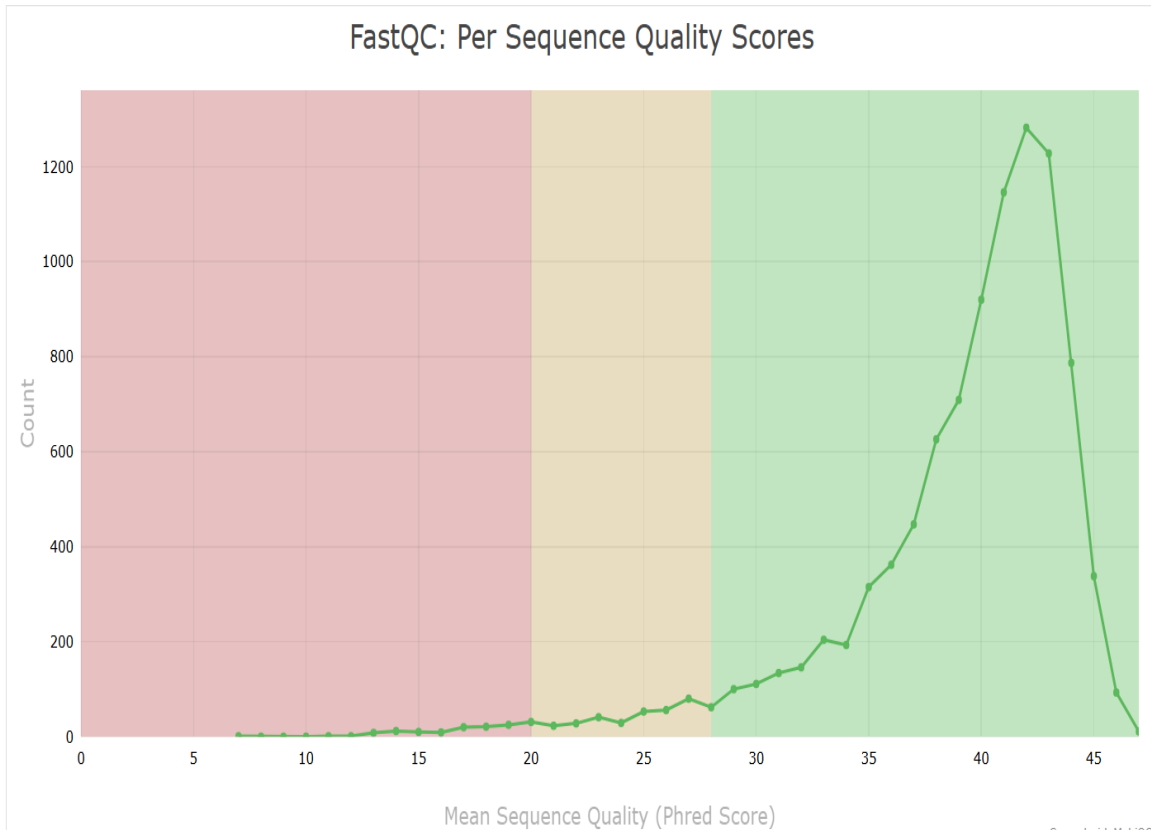
Quality collapses at read end: trim



Green along the whole read: good

This is the plot that tells you whether to trim your reads.

Per sequence quality scores



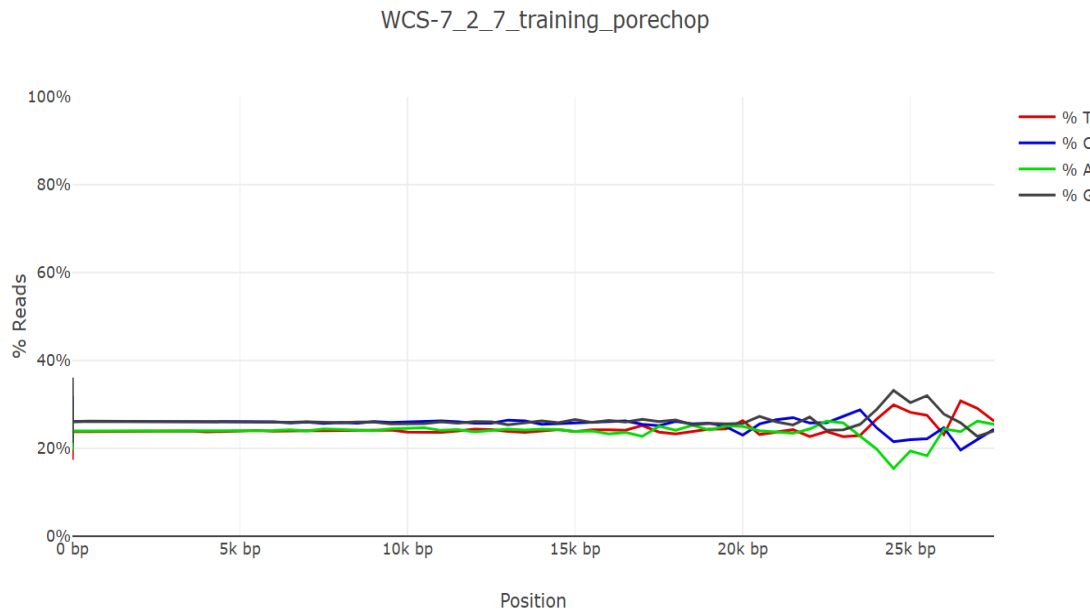
Mean quality of each read. Expect a tight peak on the right

A small low-quality tail = poorly imaged reads (edge of field of view). It should be a minor share

WARNING: most frequent mean Q < 27 (about 0.2% error)

FAIL: most frequent mean Q < 20 (about 1% error)

Per base sequence content



The bias should not go beyond the first ~10 bases

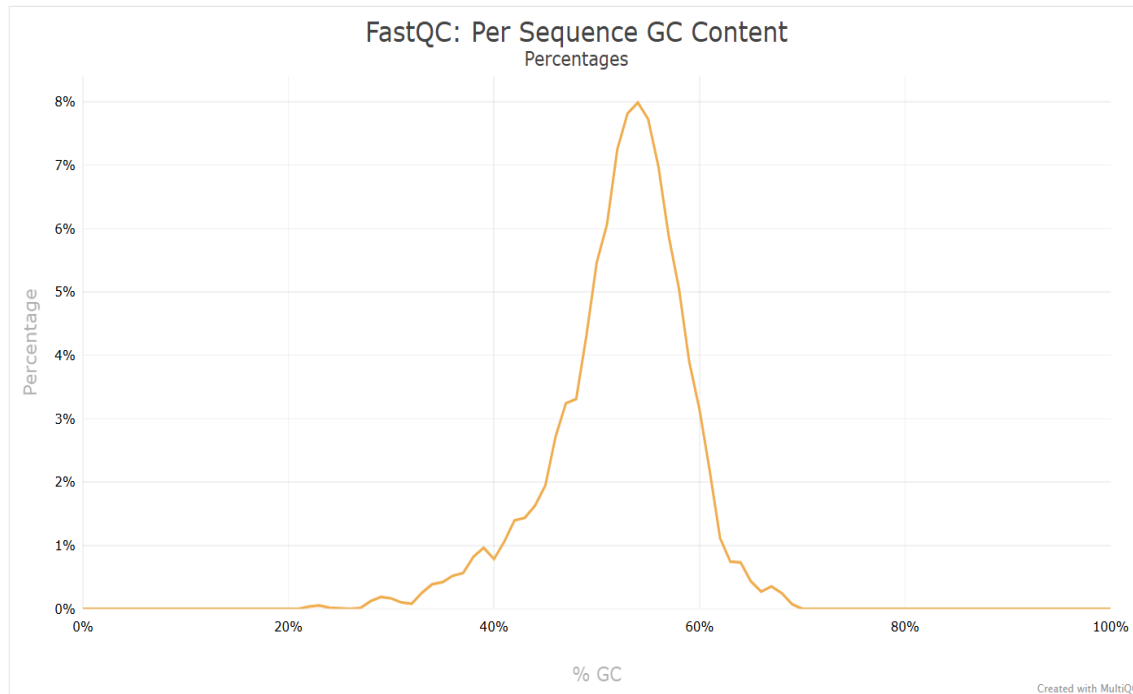
Expect flat, parallel lines with %A = %T and %G = %C

**Warn: A vs T or G vs C differ by > 10%.
Fail: > 20%**

Common reasons for a flag

- Over-represented sequences (adapters, rRNA)
- Biased fragmentation: first ~10-12 bases (random hexamers, tagmentation)
- Biased libraries (e.g. bisulfite)
- Data type: FastQC was designed for genomic data

Per sequence GC content



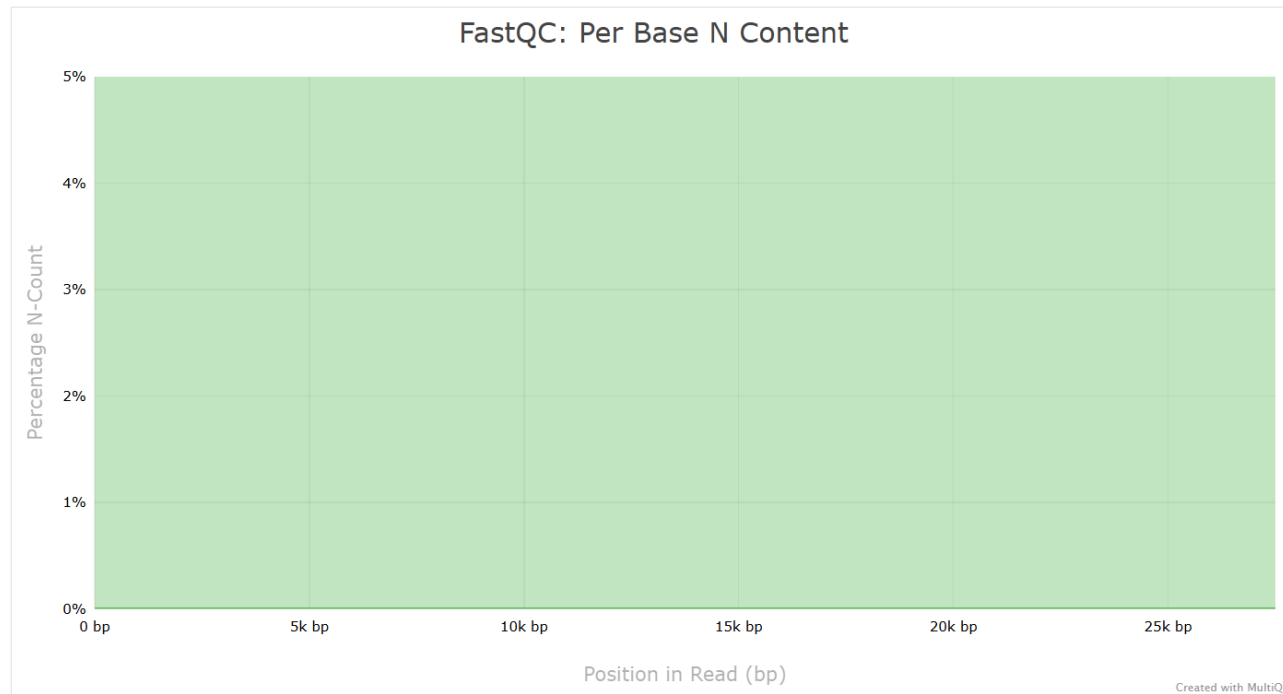
Red = observed reads. Blue = theoretical normal. The central peak matches the genome GC%

WARNING: deviations > 15% of reads

FAIL: deviations > 30% of reads

Odd shape (broad, two peaks): possible contamination (e.g. adapters) or a mixed sample. RNA-seq and metagenomics can also look unusual

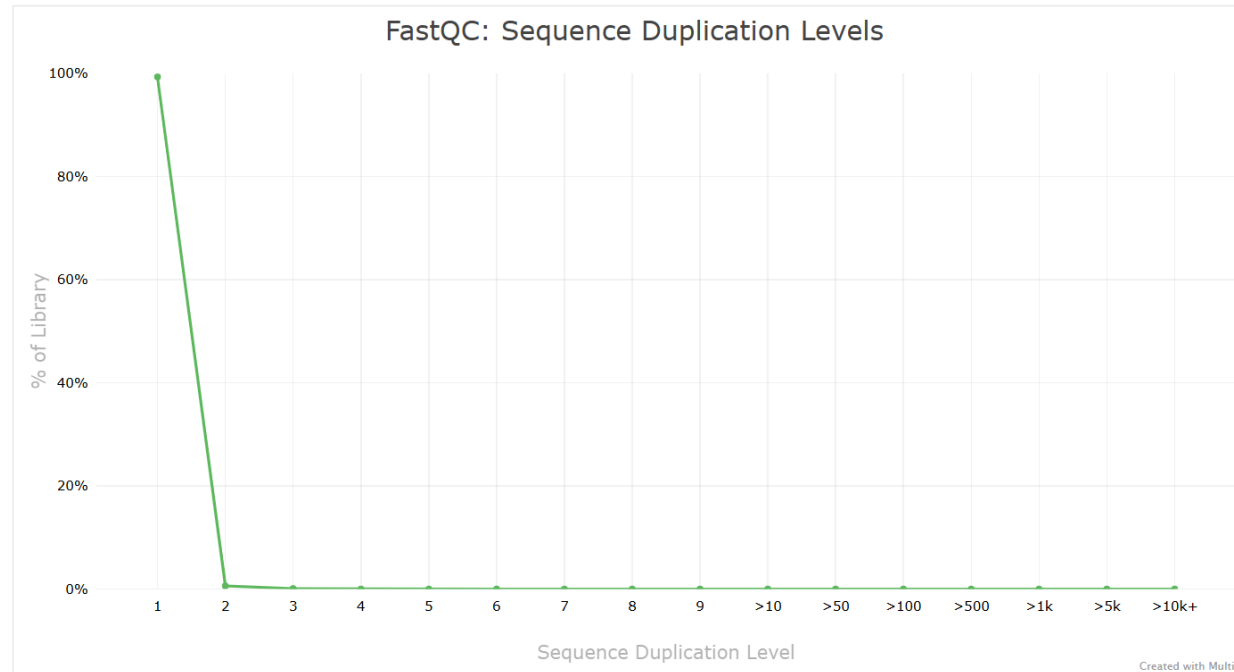
N content and read length



N content: warn > 5%, fail > 20% at any position. A few Ns near the end are normal

Length: warn if lengths differ, fail if any length is zero. Variable lengths are expected after trimming

Sequence duplication levels

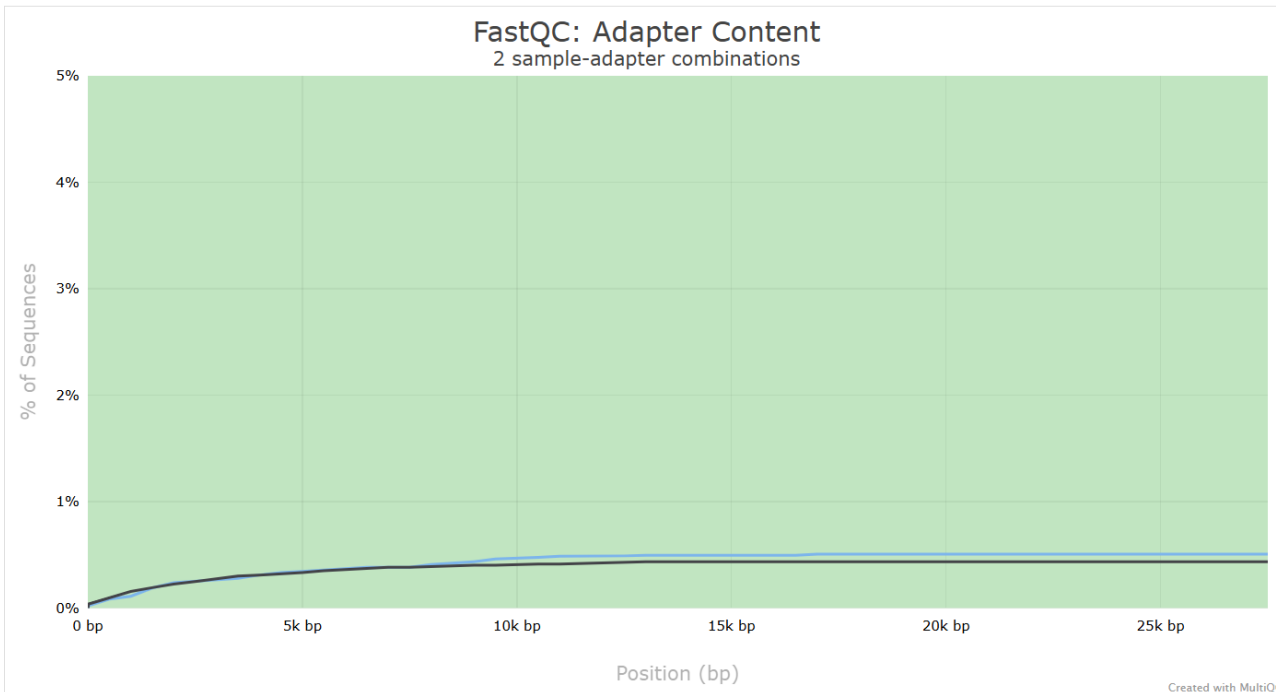


● **Diverse library: most sequences occur once (far left)**

● **Peaks on the right: enrichment bias, low-complexity contaminant or technical duplication**

Warn: > 20% non-unique | Fail: > 50% | PCR duplicates = problem; abundant RNA-seq transcripts = expected

Adapter content



Cumulative % of reads with an adapter at each position

Adapters appear when the insert is shorter than the read length

Warn: adapter in > 5% of reads

Fail: adapter in > 10% of reads

Flat at 0: already clean

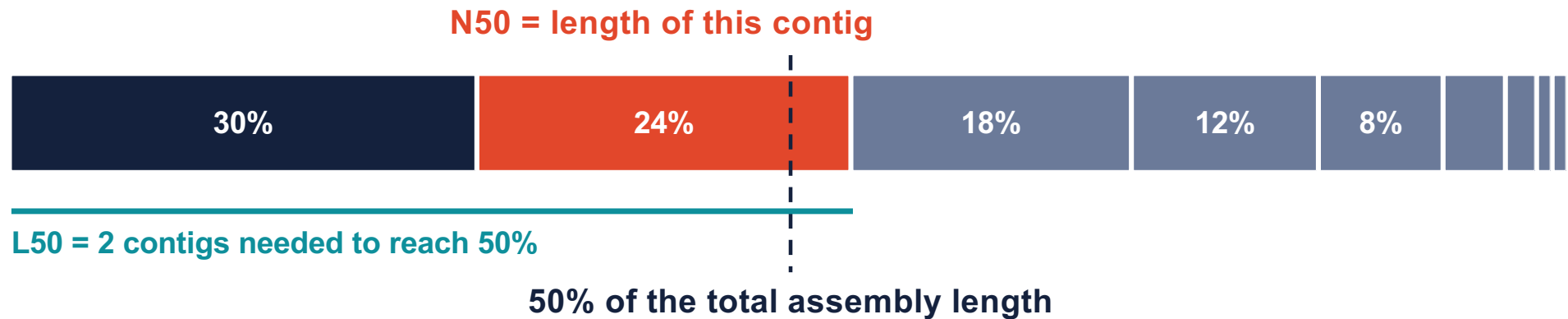
Rising: trim adapters

Early steep rise: DNA too fragmented

Use it to decide whether to trim adapters

Assembly QC (2): understanding N50

Contigs sorted by size, longest to shortest (total = 100%)



Higher N50 = less fragmented genome

Nanopore QC and multi-sample QC

Nanopore: NanoPlot

- Also available online (nanoplot.bioinf.be)
- Check the number of reads and the GC%, as for Illumina
- Do the longest reads look right? Generally > 5,000 bp
- Main difference from Illumina: quality per base

Many samples: MultiQC

- Combines results from many samples and tools in one report
- Outlier samples stand out immediately
- Run it after FastQC, and again after trimming

Trimming & Filtering Nanopore Reads



Finding and trimming adapter sequences from in ONT reads

Adapter can either be at:

- the end → trimmed off
- middle → indicate chimeric reads → split into separate reads

Takes time: aligns the reads in order to detect adapters

Supports demultiplexing for barcoded reads (Native, PCR, Rapid kits)

wdecoster/**nanofilt**

Filtering and trimming of long read sequencing data

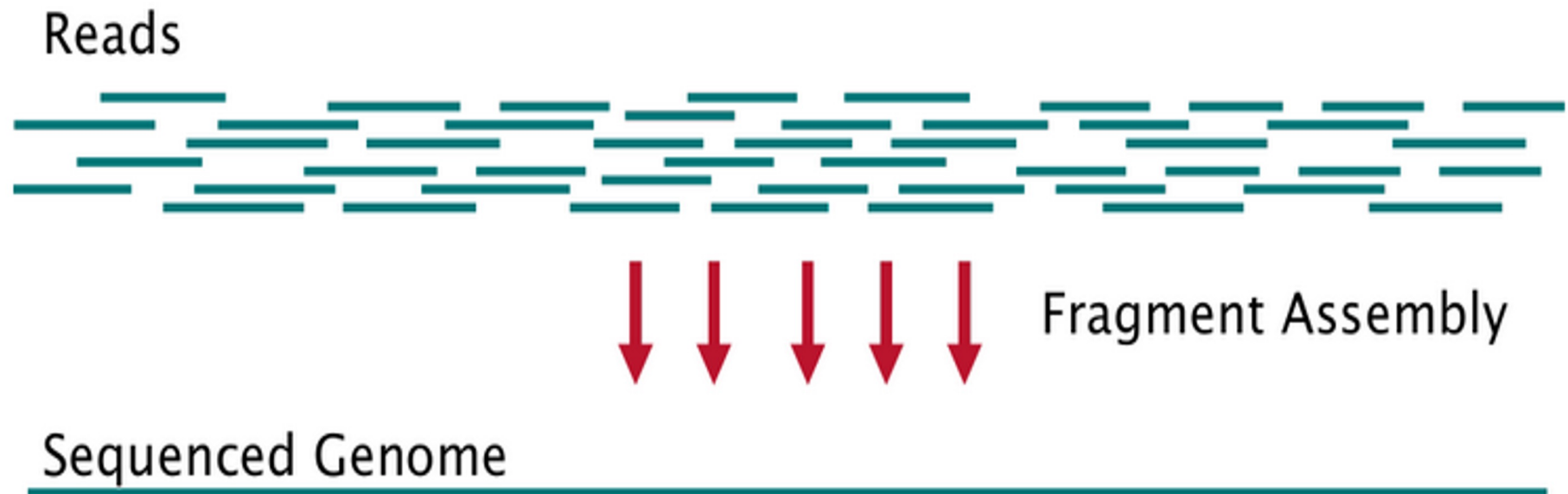


Nanofilt is a read filtering tool. It lets you:

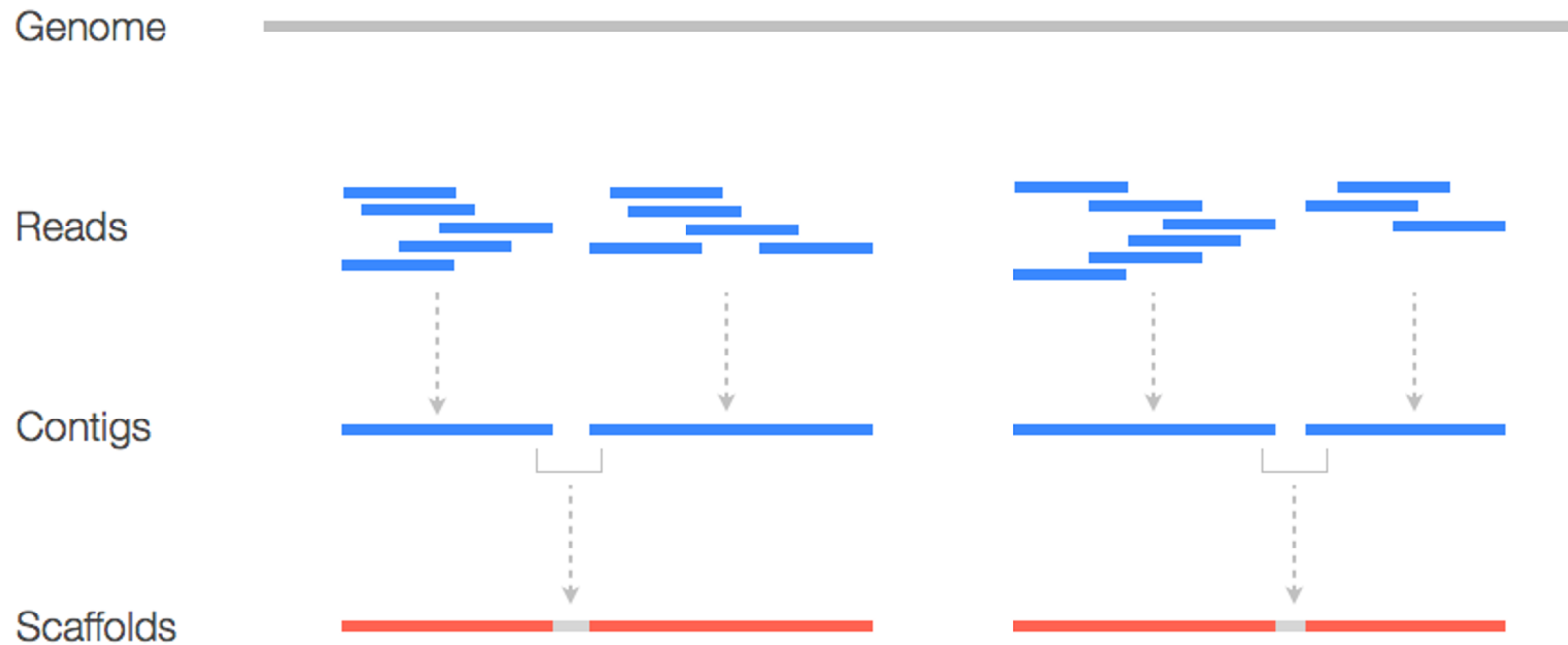
- Filter by **read length** (min/max)
- Filter by **quality score** (Q)

De Novo Assembly

Assembly in a perfect world



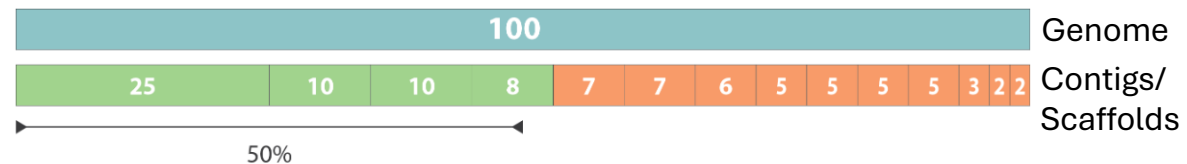
Assembly in real world



De-novo Assembly

Contigs: continuous, gapless stretches of DNA sequence.

Scaffolds: ordered arrays of contigs connected by gaps to represent a more complete view of the genome



Genome size = 100 bp

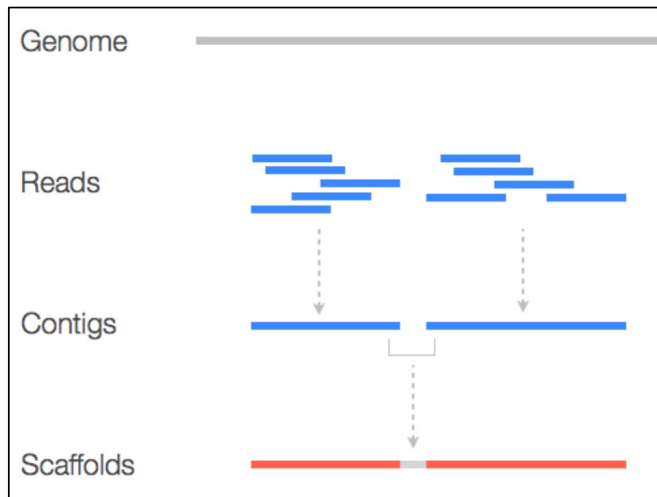
Sort the contigs/scaffolds by decreasing size.

N50: length of the shortest contig/scaffold that covers at least 50% of the genome

L50: Count until the smallest number of sequences that make 50% of genome

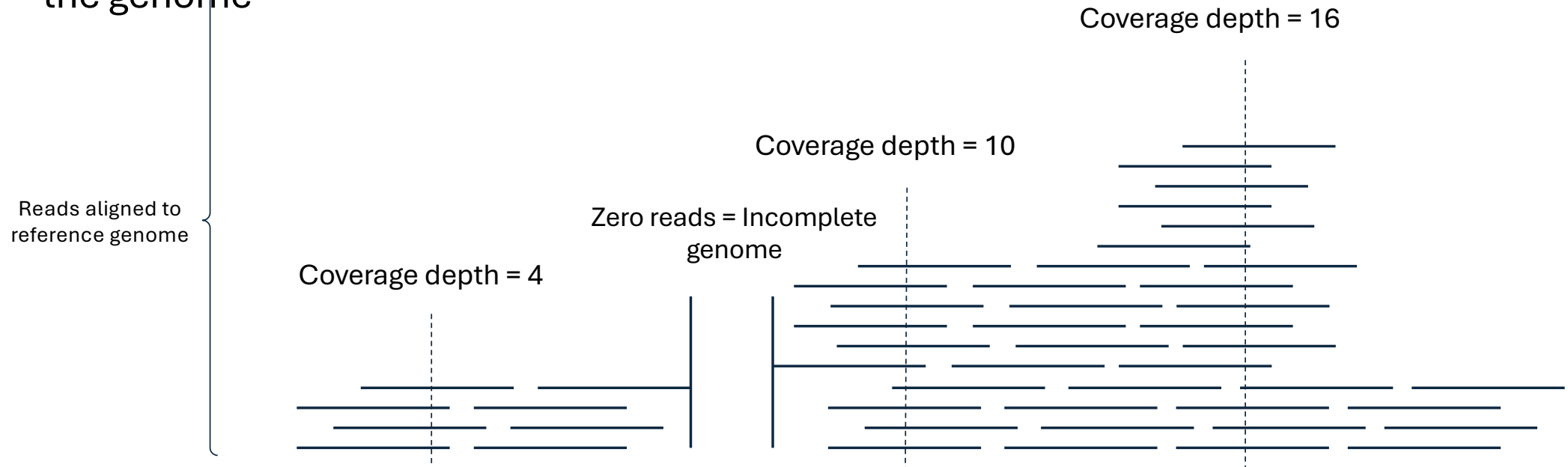
N50: 8 & L50: 4

Goal is to have high N50 and low L50.



Coverage depth/Read Depth

Coverage = How many times each base has been sequenced on average throughout the genome



Flye

- de Novo assembly = like rebuilding a shredded book from noisy strips (reads).
- Problem: repeated phrases (like “once upon a time”) confuse the reconstruction.
- Traditional assemblers (spades) need exact matches.
- Flye works with approximate matches (ONT prone to errors).
- Result: Flye builds contigs from unique regions, and shows repeats as unresolved if no read spans them fully.

Discussion

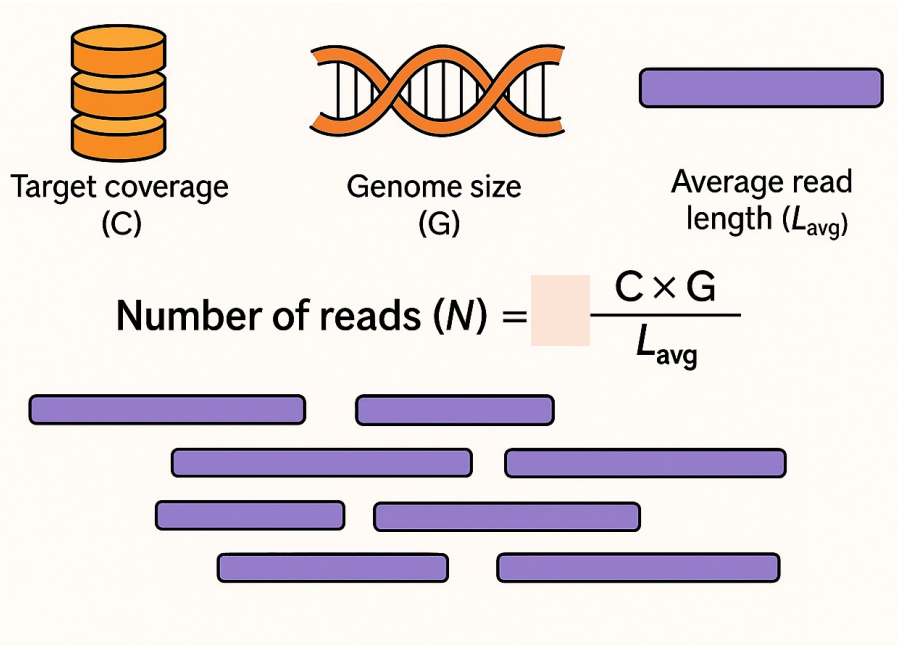
Metric	Sample A	Sample B	Sample C	Comments
Genome size (Mb)	3.8	3.8	3.8	All match the expected size
Coverage (×)	4	180	120	B best depth, C is okay but A we may have gaps
# Contigs	22,000	95	130	Fewer is better: B most contiguous, C moderate, A fragmented.
N50 (kb)	20	210	160	Longer contigs better: A very short contigs
L50	100	6	8	Highly fragmented
GC content (%)	60.5	41.5	41.5	Possible contamination on sample A

How To Determine the Number of Samples to Load

Example 1:

Organism: *Mycobacterium tuberculosis* (4.4 Mbp/4,400,000 bp)

Coverage: 100 X



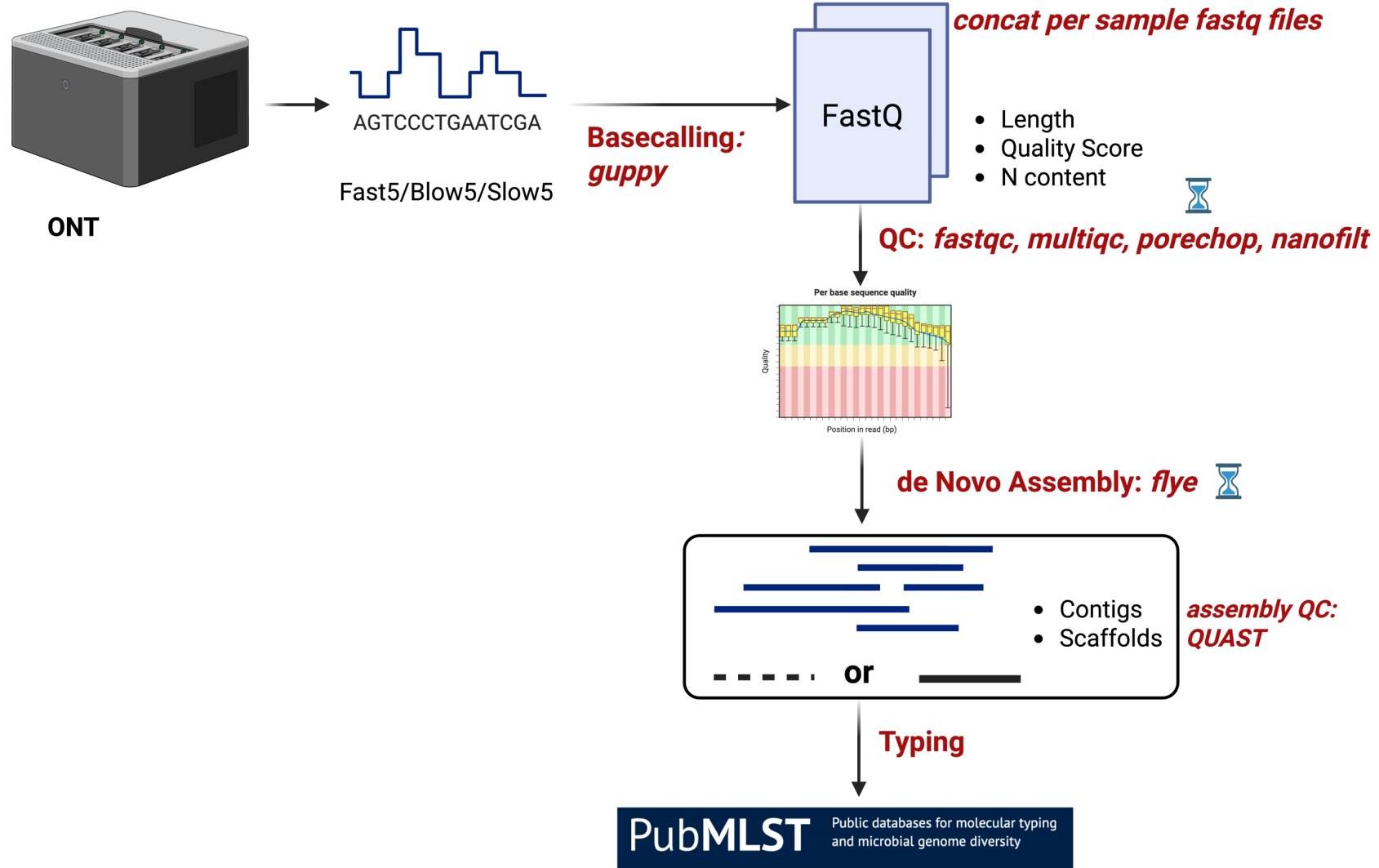
Bases per Sample (with 100X): 4.4 Mbp X 50 = 220 Mb

Average ONT Read Length: 5kbp/5,000bp

Reads per sample (5 kb reads): 220 Mb ÷ 5 kb ≈ 44,000 reads

How many TB samples will you load per run assuming a maximum of 5Gb & same read length (5kbp): 22 samples

Summary of the Pipeline



Thank you

